

Candidate Name

Centre Number

Candidate Number

Chimwera T



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY**9189/5**

PAPER 5 Practical Test

JUNE 2015 SESSION**1 hour 20 minutes**

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

TIME 1 hour 20 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 6 printed pages and 2 blank pages.

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- 1 You are to determine the percentage purity of a sample of contaminated sodium chloride.

You are provided with:

FA1 the contaminated sodium chloride sample

FA2 2.0 mol dm^{-3} hydrochloric acid

FA3 $0.098 \text{ mol dm}^{-3}$ ammonia

- (a) (i) Using a balance, measure between 3.45 g and 3.55 g of FA1 into a 100 cm^3 beaker and record your weighings in Table 1.1.

Table 1.1.

mass of beaker + FA1/g	
mass of empty beaker/g	
mass of FA1 used/g	3.50

- (ii) Using a burette add between 34.50 cm^3 and 35.50 cm^3 of FA2 into the beaker containing FA1. Record your burette readings in Table 1.2.

Table 1.2.

final burette reading/ cm^3	
initial burette reading/ cm^3	
volume of FA2 used / cm^3	35.00

- (iii) Dissolve all the FA1 by stirring with a glass rod. Transfer the resulting mixture into a 250 cm^3 volumetric flask and make up the contents to the mark using distilled water.

Label it FA4.

Pipette 25.00 cm^3 of FA3 into a conical flask and titrate it with FA4 using methyl orange as an indicator.

Repeat the titration as many times as is necessary to obtain accurate results.

Record your readings in Table 1.3.

Table 1.3.

final burette reading/cm ³			
initial burette reading/cm ³			
volume of FA4 used / cm ³	15.00	15.00	15.00

For
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Use

Summary

25.0 cm³ of FA3 required 15.00 cm³ of FA4, for a complete reaction. Show with a tick(✓) the results used to obtain this value of FA4.

(b) Calculate

(i) the number of moles of ammonia present in 25.0 cm³,

$$\begin{aligned}
 n(\text{NH}_3) &= \cancel{CV} = 0.098 \times \frac{25}{1000} \\
 &= 0.025 \text{ moles} \times 0.098 \\
 &= \underline{0.00245 \text{ moles}}
 \end{aligned}$$

[1]

(ii) the concentration of the hydrochloric acid that reacted with FA3,

$$\begin{aligned}
 \text{NH}_3 + \text{HCl} &\rightarrow \text{NH}_4^+ \text{Cl}^- \quad \text{mole ratio 1:1} \\
 n(\text{HCl}) \cdot 0.00245 &= n(\text{NH}_3) \\
 \therefore C &= \frac{n}{V} = \frac{0.00245}{\left(\frac{15}{1000}\right)} = \underline{0.163 \text{ mol/dm}^3}
 \end{aligned}$$

[2]

→ 24

4

- (iii) the number of moles of hydrochloric acid placed in the beaker.

$$n = c \times V$$

$$= \frac{35.50}{1000} \times 2.0$$

$$= 0.071 \text{ moles}$$

[2]

- (c) (i) Determine the number of moles of hydrochloric acid that reacted with FA1.

$$n(\text{HCl}) \times 5 = 0.00245 \times 10$$

$$= 0.0245$$

$$\therefore n(\text{HCl}) = 0.071 - 0.0245$$

$$= 0.0465$$

[2]

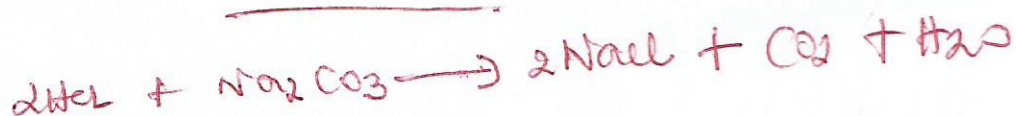
- (ii) Given that the only impurity in FA1 is Na_2CO_3 , calculate the mass of Na_2CO_3 present in FA1 used.

[M_r of sodium carbonate = 106]

$$n(\text{HCl}) = n(\text{Na}_2\text{CO}_3)$$

$$0.0465 \times \frac{1}{2} \times 106$$

$$= 2.4655$$



[2]

→ 29

5

(iii) Determine the percentage purity of sodium chloride sample.

For
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Use

$$\frac{3.5 - 2.465}{3.5} \times 100\%$$

$$= \frac{1.0355}{3.5} \times 100\%$$

$$= 29.59\%$$

[1]
[30 marks]

1

1

31

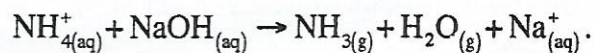
31

ASSESSMENT OF PLANNING SKILLS

DO NOT CARRY OUT THE EXPERIMENT

- 2 Sodium hydroxide displaces ammonia from its salts when the mixture is heated.

The reaction is as shown,



You are required to give a plan that will enable you to determine the percentage composition of NH_4^+ ions in an ammonium sulphate $(\text{NH}_4)_2\text{SO}_4$, sample.

Assume you are provided with the following:

a sample of solid $(\text{NH}_4)_2\text{SO}_4$,

2.5 mol dm⁻³ $\text{NaOH}_{(\text{aq})}$

50 cm³ measuring cylinder

250 cm³ conical flask

100 cm³ beaker

burner

cotton wool soaked in $\text{CaCl}_{2(\text{aq})}$

laboratory apparatus of your choice

Present your plan as a sequence of numbered steps.

The plan

[Total:10]